



ECEN 314 - Signals and Systems

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Module 7 - Stability of LTI Systems

Learning Objectives



- Be able to determine the stability of a system by observing the locations of the poles of its transfer function in the s -plane.
- Be able to use the Routh-Hurwitz procedure to determine the stability of a system.

Stability of LTI Systems



Previously we found that the output, $y(t)$, of an LTI system due to an input, $x(t)$ is given by a convolution

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(v)h(t-v)dv = \int_{-\infty}^{\infty} h(v)x(t-v)dv.$$

In the next few slides, we seek to find under what conditions such a system would exhibit stability.

Recall that a system is BIBO (bounded input, bounded output) stable if any input which is always finite produces an output which is always finite.

The magnitude of the output is

$$|y(t)| = \left| \int_{-\infty}^{\infty} x(v)h(t-v)dv \right| \leq \int_{-\infty}^{\infty} |x(v)h(t-v)| dv = \int_{-\infty}^{\infty} |x(v)||h(t-v)| dv.$$

Now suppose the input is bounded so that $|x(v)| \leq c_1$ for all t . Then

$$|y(t)| \leq c_1 \int_{-\infty}^{\infty} |h(t-v)| dv = c_1 \int_{-\infty}^{\infty} |h(w)| dw.$$

The system will be BIBO stable provided that the impulse response is absolutely integrable:

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty.$$

Stability of LTI Systems



Consider an LTI system with a general transfer function of the form

$$H(s) = \frac{(s-z_1)(s-z_2)\cdots(s-z_m)}{(s-p_1)(s-p_2)\cdots(s-p_n)}. \quad (\text{assume } m < n)$$

Furthermore, suppose all the poles are distinct (no repeated poles). Then using partial fractions, we can express the transfer function in the form

$$H(s) = \frac{A_1}{s-p_1} + \frac{A_2}{s-p_2} + \cdots \frac{A_n}{s-p_n},$$

where all of the constants A_k are finite. The corresponding impulse response will be

$$h(t) = (A_1 e^{p_1 t} + A_2 e^{p_2 t} + \cdots + A_n e^{p_n t})u(t).$$

Allowing for the possibility that some or all of the poles may be complex, write $p_k = \sigma_k + j\omega_k$. Then

$$\begin{aligned} |h(t)| &\leq \sum_{k=1}^n |A_k| |\exp(\sigma_k t + j\omega_k t)| u(t) \\ &= \sum_{k=1}^n |A_k| |\exp(\sigma_k t)| |\exp(j\omega_k t)| u(t) = \sum_{k=1}^n |A_k| \exp(\sigma_k t) u(t) \end{aligned}$$

Stability of LTI Systems



$$\begin{aligned} |y(t)| &\leq c_1 \int_{-\infty}^{\infty} |h(t)| dt \leq c_1 \int_{-\infty}^{\infty} \sum_{k=1}^n |A_k| \exp(\sigma_k t) u(t) dt \\ &= c_1 \sum_{k=1}^n |A_k| \int_0^{\infty} \exp(\sigma_k t) dt = c_1 \sum_{k=1}^n |A_k| \left[\frac{1}{\sigma_k} \exp(\sigma_k t) \right]_0^{\infty}. \end{aligned}$$

The term $\exp(\sigma_k t)$ evaluated at $t = \infty$ will be finite only if $\sigma_k < 0$. In which case we have

$$|y(t)| \leq c_1 \sum_{k=1}^n |A_k| \left[-\frac{1}{\sigma_k} \right].$$

→ We have shown that in this case, the output is bounded (by a finite #). **Note:** A similar line of reasoning can be followed if some or all of the poles are repeated.

We have proven (sort of) the following important result:

Theorem: An LTI system described by an ODE is BIBO stable if all of the poles of the transfer function are in the left half of the complex s-plane (real parts are negative).

Examples



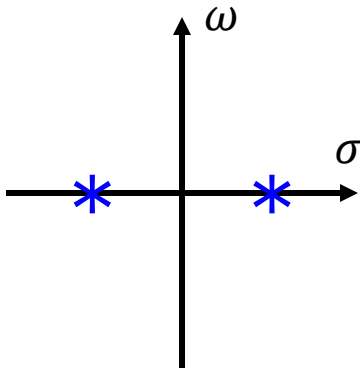
Example 7.1:

$$y''(t) - y(t) = x(t)$$

$$(s^2 - 1)Y(s) = X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2 - 1}$$
$$= \frac{1}{(s + 1)(s - 1)}$$

One pole in the RHP, one pole in the LHP → the system is not stable.



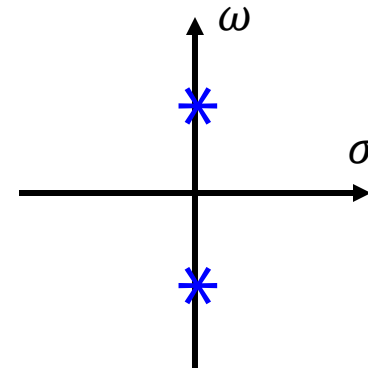
Example 7.2:

$$y''(t) + y(t) = x(t)$$

$$(s^2 + 1)Y(s) = X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2 + 1}$$
$$= \frac{1}{(s + j)(s - j)}$$

Both poles are on the imaginary axis. We call this a “marginally stable” system (marginally stable = not stable).



Determining Stability of High Order Systems



For higher order systems, finding the roots of the polynomials may be difficult. We can always resort to numerical computations, but there are some results that can determine stability without having to explicitly calculate the roots of the characteristic polynomial.

First consider a system with several real poles in the LHP. Then the characteristic polynomial will be of the form

$$b(s) = (s + p_1)(s + p_2) \cdots (s + p_n),$$

where the p_k are now all positive numbers (poles at $s = -p_k$). When we multiply this out, the resulting polynomial will have all positive coefficients.

Similarly with a pair of complex conjugate poles in the LHP ($s = -\sigma_k \pm j\omega_k$, with $\sigma_k > 0$) the polynomial will look like

$$(s + \sigma_k + j\omega_k)(s + \sigma_k - j\omega_k) = s^2 + 2\sigma_k s + \sigma_k^2 + \omega_k^2.$$

This also has all positive coefficients.

A stable system will have a characteristic polynomial that has either all positive or all negative coefficients. If the coefficients alternate signs it is necessarily unstable. If the coefficients do not alternate signs, it could still be unstable.

Bonus Material: Routh-Hurwitz Criterion



There is a procedure that goes by the name of Routh-Hurwitz that can be used to determine if a system is stable or not. It involves the creation of the RH table which can be generated following a fairly simple pattern:

Step 1: Populate the first two rows of the RH table with the coefficients of the characteristic polynomial starting with the highest order coefficient and alternately using the first two rows.

A	C
B	D
E	

Step 2: Starting in Row 3, form each new entry (E) according to

$$E = \frac{BC - AD}{B},$$

where A and B are the leftmost entries in the two rows above E and C and D are the entries from the two rows above E and one column to the right of E . Continue until all entries are 0.

Step 3: The number of roots in the RHP will be equal to the number of sign changes in the entries in the first column. → The system will be stable if all the entries in the first column are of the same sign.

Bonus Material: Routh-Hurwitz Criterion



Example 7.3: $b(s) = s^4 + 4s^3 + s^2 + 14s + 40$

	Col. 1	Col. 2	Col. 3
Row 1	1	1	40
Row 2	4	14	0
Row 3	-2.5	40	0
Row 4	78	0	0
Row 5	40	0	0

$$R3, C1: E = \frac{1 \cdot 4 - 1 \cdot 14}{4} = -2.5$$

$$R3, C2: E = \frac{4 \cdot 40 - 1 \cdot 0}{4} = 40$$

$$R4, C1: E = \frac{(-2.5) \cdot 14 - 4 \cdot 40}{-2.5} = 78$$

$$R5, C1: E = \frac{78 \cdot 40 - (-2.5) \cdot 0}{78} = 40$$

Going down Column 1, we encounter two sign changes in the entries $4 \rightarrow -2.5$ and $-2.5 \rightarrow 78$. Therefore, there should be two roots in the RHP.

In fact, the roots of this polynomial are $r_1 = -1$, $r_1 = -2$, $r_1 = 1 + j$, $r_1 = 1 - j$.

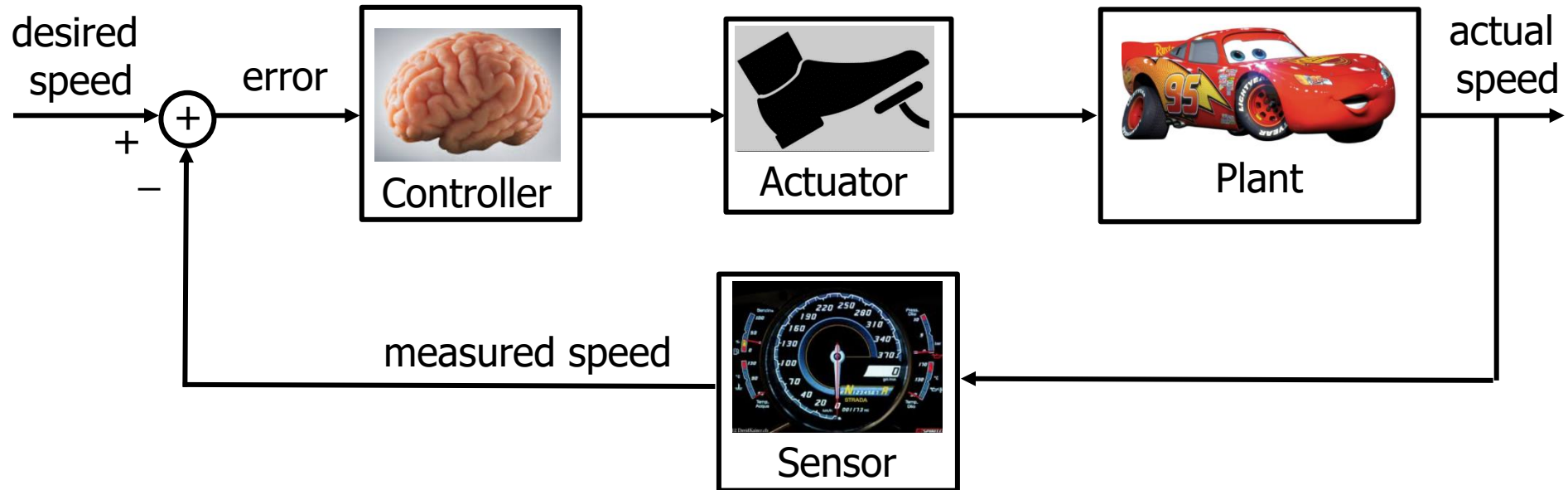
Bonus Material: Routh-Hurwitz Criterion



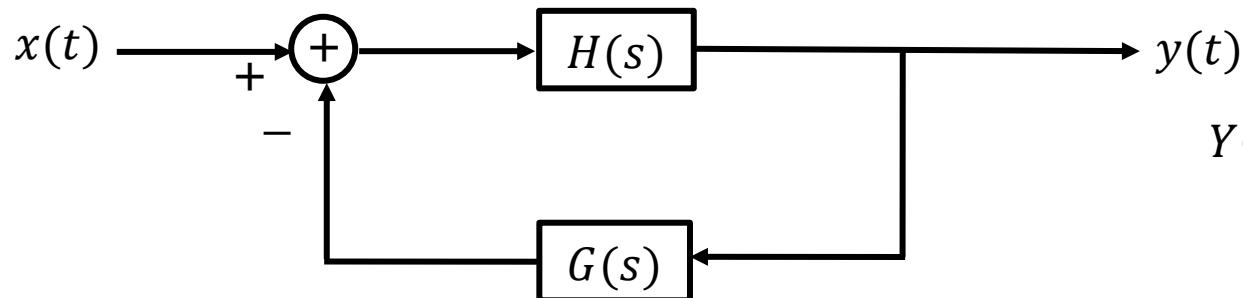
Example 7.4: Use Routh-Hurwitz to determine the number of zeros in the RHP of the polynomial $b(s) = s^4 + 10s^3 + 35s^2 + 50s + 24$

Bonus Material:

Application: Feedback Control Systems



Each block in this control system can be modelled as a system with a transfer function. To simplify the system, we will combine the Controller/Actuator/Plant into a single equivalent system



$$Y(s) = [X(s) - Y(s)G(s)]H(s)$$
$$\frac{Y(s)}{X(s)} = \frac{H(s)}{1 + G(s)H(s)}$$

Bonus Material:

Application: Feedback Control Systems



Example 7.5: Suppose in our generic feedback control system, $G(s) = 1 \rightarrow$ the speedometer perfectly and instantaneously measures the actual speed.

Furthermore, suppose $H(s) = \frac{K}{s^4 + 10s^3 + 35s^2 + 50s + 24}$. For what values of K will the feedback control system exhibit a stable behavior?

In this case, the overall system has a transfer function given by

$$\begin{aligned}\frac{Y(s)}{X(s)} &= \frac{H(s)}{1 + G(s)H(s)} \\ &= \frac{\frac{K}{s^4 + 10s^3 + 35s^2 + 50s + 24}}{1 + \frac{K}{s^4 + 10s^3 + 35s^2 + 50s + 24}} \\ &= \frac{K}{s^4 + 10s^3 + 35s^2 + 50s + 24 + K}.\end{aligned}$$

From the previous example, we know that $s^4 + 10s^3 + 35s^2 + 50s + 24$ has all of its roots in the LHP. We can use Routh-Hurwitz to help us determine which values of K will lead to stability.

Bonus Material:

Application: Feedback Control Systems



$$b(s) = s^4 + 10s^3 + 35s^2 + 50s + 24 + K$$

	Col. 1	Col. 2	Col. 3
Row 1	1	35	$24 + K$
Row 2	10	50	0
Row 3	30	$24 + K$	0
Row 4	$42 - \frac{K}{3}$	0	0
Row 5	$24 + K$	0	0
Row 6	0	0	0

Since the first 3 terms in the first column are all positive, the remaining terms must also be positive for the system to be stable.

$$42 - \frac{K}{3} > 0 \rightarrow K < 126$$

$$24 + K > 0 \rightarrow K > -24$$

The system will be stable for any K in the range

$$-24 < K < 126$$